

Şeyma Karadereli

PH.D. IN MATHEMATICS

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RESEARCH INTERESTS

Low-dimensional topology, contact topology, and singularity theory. Main focus on real algebraic realization of contact structures and Milnor open books supporting contact structures.

EDUCATION

Ph.D. in Mathematics, Boğaziçi University

2020 - Aug 2026

Advisor: Ferit Öztürk

Dissertation Title: Real Algebraic Overtwisted Contact Structures on Milnor Fillable 3-Manifolds

M.S. in Mathematics, Boğaziçi University

2017 - 2020

Advisor: Ferit Öztürk

Thesis Title: Algebraic Overtwisted Contact Structures on the 3-Sphere

B.S. in Mathematics, Boğaziçi University

2012 - 2017

ACADEMIC POSITIONS

Graduate Teaching Assistant

Jan 2020 - Present

Boğaziçi University

Visiting Ph.D. Student

Feb 2023 - Jun 2023

Erwin Schrödinger International Institute for Mathematics and Physics

Graduate Teaching Assistant

Mar 2019 - Jan 2020

Özyeğin University

PUBLICATIONS AND PREPRINTS

Real Algebraic Overtwisted Contact Structures on 3-Spheres (with F. Öztürk), *Algebraic & Geometric Topology* **25** (2025), DOI: 10.2140/agt.2025.25.1377

TALKS

Real Algebraic Overtwisted Contact Structures on 3-Spheres,

Young Women in Geometric Topology *Hausdorff Research Institute for Mathematics, Bonn*

Jul 2026

IMBM Istanbul Topology Days *Boğaziçi University, Istanbul*

May 2026

6th Austrian Day of Women in Mathematics *University of Graz*

Feb 2026

Winter School in Low-Dimensional Topology and Singularity Theory *Erdős Center, Budapest*

Jan 2023

HONORS, SCHOLARSHIPS, AWARDS

National Scholarship Program for PhD Students, TUBITAK (The Scientific and Technological Research

Council of Türkiye)

2020-2024

Junior Research Fellowship, Erwin Schrödinger International Institute for Mathematics and Physics

2023

National Scholarship Program for MSc Students, TUBITAK	2017-2019
Undergraduate Scholarship Program, TUBITAK	2014-2017
Undergraduate Scholarship Program, Government of Türkiye	2012-2014

TEACHING EXPERIENCE

Teaching Assistant Coordinator, Boğaziçi University 2022

Department of Mathematics, team of approx. 10 graduate TAs

- Distributed tasks among TAs
- Organized exam proctoring; confirmed exam dates, determined external proctor needs and communicated with the Dean's Office
- Ran orientation for incoming teaching assistants
- Served as the main point of contact between TAs and the department

Graduate Teaching Assistant, Boğaziçi University Jan 2020 - Present

Conducted weekly problem sessions, selected exercises together with the instructor and prepared written solutions to problem sets.

Held office hours, graded exams, homework and quizzes.

Courses:

- Math 101 - Calculus 1 (4 semesters)
- Math 102 - Calculus 2 (3 semesters)
- Math 201 - Matrix Theory (6 semesters)
- Math 105 - Calculus for Social Sciences (3 semesters)
- Math 111 - Introduction to Mathematical Structures (3 semesters)
- Math 162 - Discrete Mathematics (2 semesters)
- Math 131 - Calculus for a Single Variable
- Math 132 - Calculus of Several Variables
- Math 221 - Linear Algebra
- Math 231 - Advanced Calculus I (3 semesters)
- Math 325 - Matrix Groups
- Math 327 - Number Theory
- Math 471 - Topology

Graduate Teaching Assistant, Özyeğin University Mar 2019 - Jan 2020

Weekly office hours, exam gradings for the courses:

- Calculus I-II for Engineering Students
- Calculus for Social Sciences
- Discrete Mathematics

OUTREACH, MENTORING, AND SERVICE

Local Organizer, IMBM Istanbul Topology Days, Boğaziçi University 2026

Organizer, Writing Group, Boğaziçi University 2026

Regular sessions where faculty members, graduate students meet to work on their own writing projects.

Volunteer Tutor, Refugee children in Türkiye 2021-2023

Helping refugee children with mathematics lessons.

Mentor, Directed Reading Program Türkiye 2021

Mentored an undergraduate student in a reading project on classification of surfaces using Morse Theory.

Accessibility Tutor , Boğaziçi University (referred by GETEM)	2018
Provided semester long one on one instruction to a student with dyslexia taking Math 202 - Differential Equations in the Department of Mathematics.	
Organizing Team , IMAGINARY Mathematics Exhibition, Istanbul	2020
Coordinated exhibition guides; the exhibition was cancelled in its first days due to COVID-19.	

WORKSHOPS AND CONFERENCES ATTENDED

31st Gökova Geometry-Topology Conference	May 2026
<i>Gökova Geometry Topology Institute</i>	<i>Muğla, Türkiye</i>
IMBM Istanbul Topology Days	May 2026
<i>Boğaziçi University</i>	<i>Istanbul, Türkiye</i>
A Conference on Developments in Contact Topology, Mapping Class Groups, and Beyond	
Mapping Class Groups and Beyond	May 2025
<i>Middle East Technical University</i>	<i>Ankara, Türkiye</i>
A workshop to celebrate contributions of Mustafa Korkmaz	
The Semester on Singularities and Low-Dimensional Topology	Spring 2023
<i>Erdős Center</i>	<i>Budapest, Hungary</i>
Winter School in Low-Dimensional Topology and Singularity Theory, The Singular Workshop, The Low-Dimensional Workshop.	
Frontiers in Geometry and Topology Summer School and Conference	Aug 2022
<i>ICTP, The Abdus Salam International Centre for Theoretical Physics</i>	<i>Trieste, Italy</i>
Geometry-Topology Student Workshop on Khovanov Homology	May 2022
<i>Gökova Geometry Topology Institute</i>	<i>Muğla, Türkiye</i>
Braids in Low-Dimensional Topology	Apr 2022
<i>ICERM</i>	<i>Virtual Attendee</i>
Tech Topology Summer School	Jul 2021
<i>Georgia Institute of Technology</i>	<i>Remote</i>
Mini School on Singularities of Surfaces	Nov 2019
<i>Galatasaray University</i>	<i>Istanbul, Türkiye</i>
Bosphorus Real Geometry Conference	Oct 2019
<i>IMBM, Istanbul Center for Mathematical Sciences</i>	<i>Istanbul, Türkiye</i>
26th Gökova Geometry-Topology Conference	Jun 2019
<i>Gökova Geometry Topology Institute</i>	<i>Muğla, Türkiye</i>
Contact Topology Workshop	Mar 2019
<i>Hacettepe University</i>	<i>Ankara, Türkiye</i>
Topology and Geometry in Low Degrees, Workshop	Dec 2018
<i>IMBM, Istanbul Center for Mathematical Sciences</i>	<i>Istanbul, Türkiye</i>
25th Gökova Geometry-Topology Conference	Jun 2018
<i>Gökova Geometry Topology Institute</i>	<i>Muğla, Türkiye</i>